

Wilf’s inequality for numerical semigroups with four generators

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Abstract

We present a computer-assisted argument for Wilf’s inequality for numerical semigroups with four minimal generators. Preferred factorizations of an Apéry set give a finite lower ideal in three exponent coordinates. We seek a quantitative surplus over the weighted downset mean inequality, large enough to absorb the exact correction in the Apéry genus formula. Residue collisions allow at most one full-support minimal exclusion. We separate the resulting geometric cases, prove a central-box/three-horn decomposition, and combine analytic moment inequalities with seven exact finite assertions. The three-plane case is entirely analytic. Each retained finite assertion is checked by two separate implementations; interval certificates cover closed real parameter regions. We also identify dimension-independent identities and a geometric obstruction to directly extending the three-horn method beyond four generators.

1 Introduction

A numerical semigroup is a cofinite additive submonoid of $\mathbb{N} = \{0, 1, 2, \dots\}$. Its embedding dimension e is the cardinality of its minimal generating set, its multiplicity m is its least positive element, and its conductor c is the least integer such that $c + \mathbb{N}$ is contained in the semigroup. Write $n = |S \cap [0, c)|$. Its genus is $g = |\mathbb{N} \setminus S|$, so $n = c - g$. Wilf’s question [Wil78] asks whether

$$en \geq c$$

for every numerical semigroup. Fröberg, Gottlieb, and Häggkvist [FGH87] established the inequality in embedding dimension at most three. This paper addresses the next dimension.

Theorem 1.1. *Every numerical semigroup with four minimal generators satisfies*

$$W_4(S) := 4|S \cap [0, c)| - c \geq 0.$$

1.1 Foundational work

The proof builds on Zhai’s [Zha11] preferred factorizations of Apéry elements and weighted mean inequality for finite lower ideals. Hellus, Rechenauer, and Waldi [HRW18] developed the monomial initial-ideal and residue-lattice interpretation, including restrictions on the supports of minimal exclusions. The classical Apéry genus formula supplies the exact link from weighted moments to Wilf’s inequality; see also [RGS09].

Two published computer-assisted results dispose of multiplicities through 19: Bruns, García-Sánchez, O’Neill, and Wilburne [BGSOW20, Theorem 4.3], and Kliem and Stump [KS22, Proposition 6.9, published version]. The clique-tree ingredient in our geometric decomposition is classical [Gav74]; its needed finite form is proved here. Connections with three-dimensional L-shapes [AGGSL15, Cho26a] and other approaches to Wilf’s problem are discussed in Section 6.

1.2 General approach

Write $S = \langle m, a_1, a_2, a_3 \rangle$, minimally. Choose one lexicographically preferred factorization of each Apéry element by a_1, a_2, a_3 . Their exponent vectors form a lower ideal $T \subseteq \mathbb{N}^3$ of cardinality m , with labels distinct modulo m . Set

$$A = \min_i a_i, \quad w_i = a_i/A, \quad M = c + m - 1, \quad H = M/A, \quad s = \sum_{x \in T} x.$$

The genus formula gives

$$mW_4(S) = A(3mH - 4w \cdot s) - m(m - 1).$$

Thus the weighted mean baseline $3mH - 4w \cdot s \geq 0$ does not suffice: the proof needs a positive surplus that pays for the correction term.

We pursue this surplus through geometry and residue collisions. A minimal excluded vector and its representative have disjoint supports, so there is at most one minimal exclusion with all coordinates positive. An ideal without such an exclusion is a central anchored box with at most three monotone rectangular horns. An ideal with one is obtained from such a completion by deleting a single upper orthant. This decomposition makes additive extremal bounds accessible to finite, exact dynamic programming.

For small multiplicity, an analytic reduction places every hypothetical counterexample inside an explicit bounded generator box. For large multiplicity, the degree of the possible positive corner and the degree of T give an exhaustive partition. The corner $(1, 1, 1)$ is handled by a counting and projection argument. Other cases use local centroid gains, axis witnesses, or interval domination of all real weights in a compact failure region. A finite degree strip also yields a continuous moment gap by averaging translated lattice meshes.

The weakest retained conclusion is $3mH - 4w \cdot s \geq m - 29/10$ for the branches not already settled directly. For $m \geq 30$, the exact moment identity and $A \geq m + 1$ then give $W_4 > -1$; integrality gives $W_4 \geq 0$.

1.3 Organization

Section 2 gives the proof and its seven finite assertions. Section 3 supplies three longer analytic arguments integral to that proof. Section 4 specifies the computational artifact and its checks. The final two mathematical sections discuss extensions beyond four generators and related work. We conclude with an account of how the proof was constructed using AI and exact computation.

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2 Proof of the four-generator inequality

All vector comparisons are coordinatewise. A lower ideal contains every nonnegative integer vector below each of its points. A positive corner means a minimal excluded point with all three coordinates positive.

2.1 Apéry representatives, moments, and collisions

The Apéry set

$$\text{Ap}(S, m) = \{w \in S : w - m \notin S\}$$

consists of the least semigroup element in each residue modulo m . It has m elements, every factorization of which avoids m . Minimality of the generators gives $0, a_1, a_2, a_3 \in \text{Ap}(S, m)$, with the generator factorizations unique. Its largest element is

$$M = c + m - 1 :$$

every $w - m$ is at most the largest gap $c - 1$, while $c - 1 + m$ is Apéry.

For each Apéry element choose the lexicographically least exponent vector $x \in \mathbb{N}^3$ with label $\lambda(x) = a \cdot x = w$. Let T be the chosen vectors. This is the preferred-factorization construction of Zhai, with the initial-ideal interpretation of Hellus–Rechenauer–Waldi [Zha11, HRW18].

To prove lower closure, take $y \leq x \in T$ and $v = x - y$. If $\lambda(y) - m \in S$, then $\lambda(x) - m \in S$, a contradiction. Thus $\lambda(y)$ is Apéry. A lexicographically earlier factorization y' would give $y' + v <_{\text{lex}} x$ of the same label. Consequently $y \in T$. In particular,

$$|T| = m, \quad 0, e_1, e_2, e_3 \in T, \quad \lambda : T \longrightarrow \text{Ap}(S, m) \text{ is bijective,}$$

and the labels on T are distinct modulo m .

Let $s = \sum_T x$, $w = a/A$, $H = M/A = \max_T w \cdot x$, and $R = \max_T |x|_1$. Every $w_i \geq 1$, so $R \leq H$. Writing the least representative of residue r as $r + k_r m$ and counting its k_r gaps proves the classical genus formula (also [Zha11, Lemma 1])

$$g = \frac{1}{m} \sum_{\text{Ap}} w - \frac{m-1}{2}.$$

Since $W_4 = 3c - 4g$, substitution gives the exact identity

$$\boxed{mW_4 = AD_0 - m(m-1), \quad D_0 = 3mH - 4w \cdot s.} \quad (2.1)$$

A minimal excluded point, or *corner*, is $p \notin T$ with $p - e_i \in T$ whenever $p_i > 0$. Let $\rho(p) \in T$ have its residue. If both $p_i, \rho(p)_i > 0$, subtracting e_i gives distinct included points of equal residue, impossible. Likewise two distinct corners sharing a positive coordinate cannot have the same residue. Hence a positive corner has representative zero, and there is at most one positive corner. All other corners have a zero coordinate. These are the support/collision consequences of [HRW18, Proposition 2.6].

2.2 Coordinate lines and two centroid bounds

For each of the three coordinate directions partition T into its nonempty coordinate lines. Use an *indexed* line family: intersecting geometric lines in different directions remain distinct. A line of length L_ℓ starts at coordinate zero and has top t_ℓ . Arithmetic progression sums give

$$\sum_\ell L_\ell = 3m, \quad \sum_\ell L_\ell t_\ell = 4s. \quad (2.2)$$

In a fixed direction its varying-coordinate contribution is twice that coordinate's moment; the other coordinates contribute their moments once. Thus, for any positive weights and any allowance $Q \geq \max_T w \cdot x$,

$$D_Q := 3mQ - 4w \cdot s = \sum_\ell L_\ell (Q - w \cdot t_\ell) \geq 0. \quad (2.3)$$

This is the dimension-three weighted downset baseline of [Zha11, Lemma 3].

Define

$$\rho_2(t) = \max\{|v|_1 : v \in \mathbb{N}^3, |v|_1 \leq 2, t + v \in T\}, \quad U_2 = \sum_\ell L_\ell \rho_2(t_\ell).$$

Choose a maximizing destination for each line. The total destination mass is $3m$, and its moment is $4s + \beta$, with $\beta \geq 0$ coordinatewise and $|\beta|_1 = U_2$. Each destination has weight at most Q . For weights at least one this proves

$$D_Q \geq U_2. \quad (2.4)$$

Only ten offsets are possible: zero, three units, three doubled units, and three sums of distinct units.

More generally, any nonnegative point masses α_x on T with total $3m$ and residual $\beta = \sum_x \alpha_x x - 4s \geq 0$ give $D_Q \geq |\beta|_1$. Indeed $D_Q = \sum_x \alpha_x (Q - w \cdot x) + w \cdot \beta$. This identity also proves the explicit exceptional witnesses below.

If $q_i = \max_T x_i > 0$, $q_* = \max_i q_i$, put

$$G = q_* \left(3m - 4 \sum_i s_i / q_i \right). \quad (2.5)$$

When $C = 3m - 4 \sum_i s_i / q_i \geq 0$, assign mass $4s_i / q_i$ to $q_i e_i$, and mass C to a longest-axis endpoint. The total is $3m$, with residual $C q_* e_j \geq 0$, proving $D_Q \geq G$. When $C < 0$, (2.3) gives $D_Q \geq 0 > G$. In all cases $D_Q \geq \max(U_2, G)$.

2.3 The final-window projection inequality

Let π_j delete coordinate j , and put

$$P(T) = \sum_j |\pi_j T|, \quad Z = \{x \in T : M - \lambda(x) < m\}, \quad K = \text{Max}(T).$$

The set Z is nonempty. Its points are maximal, since an included successor would increase their labels by $a_j > m$ beyond M . For a nonempty point set X , define

$$E(X) = \sum_X |x|_1 - \sum_j \max_X x_j, \quad \Phi(T, X) = P(T) - 3|X| - E(X).$$

Here $E(X) \geq 0$. Adding a point increases each coordinate sum at least as much as its coordinate maximum, so $E(X) \leq E(Y)$ for $X \subseteq Y$. Consequently $\Phi(T, Z) \geq \Phi(T, K)$.

We prove

$$\boxed{mW_4 \geq m\Phi(T, Z) + E(Z)}. \quad (2.6)$$

Use the unnormalized weights a in the line formula, and write $M - \lambda(t_\ell) = md_\ell + \beta_\ell$, $0 \leq \beta_\ell < m$. For a direction- j line of length L , define

$$R_j(L) = \sum_{r=0}^{L-1} [ra_j]_m, \quad C_\ell = \#\{r < L : \beta_\ell + [ra_j]_m \geq m\}, \quad \sigma_\ell = mLd_\ell + mC_\ell - R_j(L).$$

Reading the line backward gives

$$L(M - \lambda(t_\ell)) = \sum_{x \in \ell} [M - \lambda(x)]_m + \sigma_\ell.$$

Each direction partitions all m residues. Summing and using (2.1) gives $mW_4 = \binom{m}{2} + \sum_\ell \sigma_\ell$.

A deep line, whose top is outside Z , has $d_\ell \geq 1$. Its positive axis-prefix residues are distinct nonzero residues on T ; therefore $R_j(L) \leq m(L-1) - \binom{L}{2}$ and $\sigma_\ell \geq m$. There are $P(T) - 3|Z|$ deep lines, because every $z \in Z$ tops exactly one line in each direction.

On a final line $d_\ell = 0$, so $\sigma_\ell \geq -R_j(L)$. Across final lines each positive axis-prefix point appears once, with exactly $E(Z)$ repeated occurrences: its multiplicity in direction j is $\#\{z \in Z : z_j \geq r\}$. Distinct axis-prefix points have distinct nonzero residues. The sum $\binom{m}{2}$ pays for one copy of each used residue; each repeated copy costs at most $m-1$. Thus

$$mW_4 \geq m(P(T) - 3|Z|) - (m-1)E(Z),$$

which is (2.6). In particular $\Phi(T, K) \geq 0$ suffices to prove Wilf.

2.4 Phase and rectangular thickening

Normalize by attained height: $u_i = w_i/H$, $v = \min_i u_i$, $U = \max_i u_i$, $\sigma = \sum_i u_i$, $\mu_i = \mathbb{E}_T(u_i X_i)$, and $\kappa = 3 - 4 \sum_i \mu_i = D_0/(mH)$. Units give $u_i \leq 1$. The mean slack of direction j line tops is defined as follows. For $x \in T$, let $t_j(x)$ be the top of the direction- j line through x , and put

$$\delta_j(x) = 1 - u \cdot t_j(x), \quad \bar{\delta}_j = \frac{1}{m} \sum_{x \in T} \delta_j(x).$$

Thus $\bar{\delta}_j$ is the length-weighted mean over direction- j lines, and the line moment identities give

$$\bar{\delta}_j = (1 + \kappa)/4 - \mu_j, \quad \sum_j \bar{\delta}_j = \kappa \geq 0.$$

Choose coordinate k of weight v . Every interval of length h has sawtooth integral

$$\int_I [r]_q dr \geq \frac{qh^2}{2(h+q)}.$$

To prove this, write $h = nq + r_0$: n full periods contribute $nq^2/2$, the remaining arc at least $r_0^2/2$; Cauchy–Schwarz gives $(nq^2 + r_0^2)/2 \geq h^2/(2(n+1))$, and $n+1 \leq (h+q)/q$.

On a coordinate- k line of length L , write $\delta_j(t)$ for the slack at its point with k -coordinate t . For $j \neq k$, these slacks have residues $[\theta - tv]_{u_j}$. Nonnegative slack and $[\theta - tv + r]_{u_j} \leq \delta_j(t) + r$, $0 \leq r \leq v$, bound the integral over the concatenated intervals of total length Lv . Since $Lv \leq 1 + v$, division gives

$$\text{mean}_{\text{line}} \delta_j \geq \frac{u_j v L}{2(1 + v + u_j)} - \frac{v}{2}.$$

Length-weighted averaging uses $v \sum L^2/m = 2\mu_k + v$, hence

$$2\bar{\delta}_j(1 + v + u_j) \geq 2u_j\mu_k - v(1 + v).$$

Sum over $j \neq k$, substitute $\mu_k = (1 + \kappa)/4 - \bar{\delta}_k$, and use $U \leq \sigma - 2v$. The remaining slack coefficient is the middle weight minus $1 + v$, nonpositive because that weight is at most one. Discarding it gives

$$\boxed{\sigma(1 - 3\kappa) \leq 4\kappa + 5v - 3\kappa v + 4v^2.} \quad (2.7)$$

If $H > 3$ and $D_0 < m$, then $0 \leq \kappa < v = 1/H < 1/3$. The right side divided by $1 - 3\kappa$ is strictly increasing in κ , with derivative numerator $4 + 12v + 12v^2 > 0$. Writing $B = \sum_i w_i$ proves the necessary failure cutoff

$$\boxed{B < 9 + 28/(H - 3).} \quad (2.8)$$

Thicken T into equal-volume half-open boxes

$$K_T = \bigcup_{x \in T} \prod_i \left[\frac{w_i x_i}{H + B}, \frac{w_i (x_i + 1)}{H + B} \right).$$

It is a positive-volume downset in the unit simplex. Its mean coordinate sum is $(w \cdot s/m + B/2)/(H + B)$, giving

$$\boxed{\kappa_c(K_T) := 3 - 4\mathbb{E}_{K_T}|Y|_1 = \frac{D_0/m + B}{H + B}.} \quad (2.9)$$

The complement is the union of upper orthants with vertices $(w_i p_i / (H + B))_i$, for the discrete minimal exclusions p . Positive diagonal scaling preserves their supports and minimality. The same calculation with an allowance Q in place of H replaces D_0 by D_Q in the identity. A continuous lower bound $\kappa_c \geq \gamma$ therefore gives

$$D_0/m \geq \gamma H - (1 - \gamma)B, \quad D_0 < m \implies \gamma H < 1 + (1 - \gamma)B. \quad (2.10)$$

Lemma 2.1 (Height cutoff). *Suppose $H > 3$, $D_0 < m$, and*

$$\frac{D_0}{m} \geq \alpha H - \beta B, \quad \alpha > 0, \quad 0 \leq \beta \leq 3.$$

Then

$$H < C := 3 + \frac{1 + 9\beta}{\alpha}. \quad (2.11)$$

Proof. The assumed moment bound gives $\alpha H < 1 + \beta B$. Substitute (2.8) and multiply by the positive quantity $H - 3$:

$$\alpha H^2 - (3\alpha + 1 + 9\beta)H + 3 - \beta = \alpha H(H - C) + 3 - \beta < 0.$$

If $H \geq C$, the left side is nonnegative because $\alpha > 0$ and $3 - \beta \geq 0$, a contradiction. $\square$

Branch	α	β	Bound
No full-support corner	1/3	2/3	$H < 24$
Short corner, high height	1/3	4/3	$H < 42$
Residual high height	5/42	37/42	$H < 78$

Table 1: Parameters for the common height-cutoff argument. The interval certificates cover the corresponding closed supersets.

2.5 Structural geometry and a reusable exact interval method

Theorem 3.1, proved in Section 3.1, proves that every finite no-full-corner lower ideal U is a disjoint central anchored box $[0, z]$ and at most three outward horns. Horn i consists of rectangles at levels $t > z_i$, with transverse caps at most z_j, z_k , nonincreasing with t . The proof constructs a chordal compatibility graph, proves its needed clique-tree property, and uses the median of three coordinate maximizers.

For a unique positive corner p , deleting its generator from the upper-ideal complement gives a finite no-full-corner U with

$$T = U \setminus (p + \mathbb{N}^3). \quad (2.12)$$

Pure-axis exclusions remain, so U is finite and has the same axes. Clipping the center and horns preserves their disjointness.

Here is the exact upper-bound method used in the three interval checks. At scale $q = 4096$, integer endpoints $L = (B_0, C_0, Q_0)$, $V = (B_1, C_1, Q_1)$ enclose a closed real parameter box for (b, d, Q) . Put

$$\ell = (q, B_0, C_0), \quad u = (q, B_1, C_1), \quad N_i = \lfloor Q_1 / \ell_i \rfloor, \quad \psi(x) = 4u \cdot x + q - 3Q_0.$$

Every relevant retained point satisfies $\ell \cdot x \leq Q_1$. Every axis of U survives clipping, so $U \subseteq \prod_i [0, N_i]$. Pointwise monotonicity gives, for every real parameter in the box,

$$q(m - (3mQ - 4w \cdot s)) \leq \sum_T \psi(x). \quad (2.13)$$

For an inclusive box $[a, d]$, its count is $\prod_i (d_i - a_i + 1)$ and twice its objective moment is that count times $\sum_i u_i(a_i + d_i)$. For a clipped box subtract the upper box $[\max(a, p), d]$. The retained feasibility maximum is

$$\max_{a_i \leq \min_i(d_i, p_i - 1)} \left(\ell_i \min(d_i, p_i - 1) + \sum_{j \neq i} \ell_j d_j \right). \quad (2.14)$$

This maximizes over the parts with some coordinate below p . An independent formula partitions by the *first* such coordinate, summing disjoint boxes with earlier coordinates at least their p_j . Their maximum heights and summed moments give the same statistics. Without clipping use the full box and its upper-corner height.

Let $C_i(t, r, s)$ be the exact section score, and call it feasible when its retained maximum height is at most Q_1 . For a possibly empty nested horn use Lemma 3.5, terminating beyond its axis cap. Enumerate every feasible retained central box and add its score to $\sum_i F_i(z_i + 1; z_j, z_k)$. The structural theorem then proves an upper bound $V_p(L, V)$ on (2.13). Independent horns may enlarge the class; they never decrease its upper bound.

Order tightening replaces endpoints by

$$L^\# = (B_0, \max(B_0, C_0), \max(B_0, C_0, Q_0)), \quad V^\# = (\min(B_1, C_1, Q_1), \min(C_1, Q_1), Q_1). \quad (2.15)$$

Its ordered intersection is unchanged; inversion proves emptiness. Splitting a tightened box at an interior integer wall yields two closed children sharing that wall and covering the parent. A finite tree reaching every node exactly once and ending only at checked empty, analytic, or exact-DP leaves covers all real root parameters.

2.6 Small multiplicity

The two published computer-assisted results [BGSOW20, Theorem 4.3] and [KS22, Proposition 6.9] settle all numerical semigroups of multiplicity at most 18 and exactly 19, respectively, without an embedding-dimension restriction.

For $20 \leq m \leq 29$ we need an analytic counterexample reduction. Theorem 3.6, proved in Section 3.2, proves Wilf whenever $T = \{x : a \cdot x \leq M\}$, including an analytic residue obstruction to its formal six-column case.

If the ideal is not full weighted, choose an omitted point of weight at most M of least degree. It is a minimal excluded point p with $a \cdot p \leq M$. Its direction- i predecessor line, for $p_i > 0$, has top $p - e_i$ and length exactly p_i . Their contribution to the unnormalized line slack is

$$\sum_{i: p_i > 0} p_i(M - a \cdot p + a_i) = M + (|p|_1 - 1)(M - a \cdot p) \geq M.$$

All other slacks are nonnegative. Therefore $W_4 < 0$, which implies the integer bound $W_4 \leq -1$, forces

$$\boxed{M \leq 3mM - 4a \cdot s \leq m(m - 2)}. \quad (2.16)$$

Every generator a_i is Apéry and at most M . After sorting, any counterexample thus belongs to the *inclusive* box

$$m < a_1 < a_2 < a_3 \leq B_m := m(m - 2).$$

Finite lemma 2.2 (Generator-box). For each $20 \leq m \leq 29$, every increasing triple in this box with $\gcd(m, a_1, a_2, a_3) = 1$ and all four generators minimal has either largest Apéry element $M > B_m$, or $W_4 \geq 0$.

Every raw triple is classified by successive exact tests: $a_1 \in \langle m \rangle$, $a_2 \in \langle m, a_1 \rangle$, $a_3 \in \langle m, a_1, a_2 \rangle$, then a nontrivial gcd, or validity. Larger generators cannot make an earlier one redundant. One path computes exact residue shortest distances d_r , adjoining a generator by two forward passes around each residue cycle. A shortest extension uses fewer than the cycle length many new copies; two passes propagate every possible starting residue around the cycle. The second path uses increasing ordinary membership: $I_j(n) = I_{j-1}(n) \vee (n \geq a_j \ \& \ I_j(n - a_j))$. It tests the complete window $[B_m - m + 1, B_m]$, which is full exactly when $M \leq B_m$. A full window and closure under $+m$ prove every larger integer is included, so gaps and conductor are fully determined.

The generator-box finite lemma contradicts (2.16) for any negative example. Thus all multiplicities through 29 are settled.

2.7 Large multiplicity: exhaustive routing and the three-plane case

Now $m \geq 30$. If there is no positive corner use Section 2.8. Otherwise let p be the unique positive corner. If $|p|_1 = 3$, then $p = (1, 1, 1)$ and the following analytic argument applies. Degree four gives a permutation of $(2, 1, 1)$, handled in Section 2.9. For $|p|_1 \geq 5$, degree $R \leq 6$ uses Section 2.10; otherwise $R \geq 7$ uses Section 2.11. These are disjoint exhaustive alternatives. Minimality gives $|p|_1 - 1 \leq R$; consequently Section 2.10 has $5 \leq |p|_1 \leq 7$, while Section 2.11 has $H \geq R \geq 7$. All coordinate permutations are applied simultaneously to weights.

Suppose $p = (1, 1, 1)$. Every point of T lies in a coordinate plane. A mixed corner q in the ij -plane has representative γe_k , $1 \leq \gamma \leq h_k = \max_T x_k$. The included point $q - e_i - e_j$ has residue $(\gamma + 1)a_k$, since $a \cdot p \equiv 0$. If $\gamma < h_k$ it collides with $(\gamma + 1)e_k$. Thus all mixed corners in that plane have representative $h_k e_k$; the collision rule permits at most one.

Here is a purely geometric counting proof of $\Phi(T, K) \geq 0$ for $m \geq 30$. Write $h = h_1 + h_2 + h_3 \geq 3$. Each plane is its full rectangle or that rectangle with one upper-right quadrant deleted; it contains $(1, 1)$. Its frontier has one of the following forms, allowing transpose:

Type	Mixed maxima	Mixed point count c_{ij}	Correction β_{ij}	Axis charge ν_{ij}
Rectangle	(h_i, h_j)	$h_i h_j$	0	0
Split, $1 \leq b < h_i$, $1 \leq d < h_j$	$(b, h_j), (h_i, d)$	$b h_j + d h_i - b d$	$b + d$	0
Side cap at i , $1 \leq b < h_i$	(b, h_j)	$b h_j$	b	h_i

Let s_f count split planes, I the globally maximal axis directions, $J = \sum_{i \in I} h_i$, $B = \sum \beta_{ij}$, $C = \sum (c_{ij} - \beta_{ij})$, and $N = \sum \nu_{ij}$. Every mixed planar maximum is globally maximal. An axis top is globally maximal exactly when both incident planes side-cap that axis. Each side-cap plane can cap only one axis, while a split plane caps none. Consequently $2|I| \leq 3 - s_f$, so $s_f + |I| \leq 3$ and $k = |K| = 3 + s_f + |I| \leq 6$. For a split, $c_{ij} - \beta_{ij} = b d + b(h_j - d - 1) + d(h_i - b - 1) \geq b + d - 1$. For a side cap $c_{ij} - \beta_{ij} \geq 0$ and $\beta_{ij} \leq \nu_{ij}$. Hence

$$B \leq C + s_f + N, \quad N \geq 2J.$$

Direct origin/axis/plane counting yields

$$m = 1 + h + B + C, \quad P(T) = m + h + 2, \quad E(K) = h + B - N + J.$$

Set $Q_* = h + C + N - J$. Then $\Phi(T, K) = Q_* + 3 - 3k$. If this were negative, integrality would give $Q_* \leq 3k - 4 \leq 14$, and

$$m = 1 + Q_* + B - N + J \leq 1 + 2Q_* - h + s_f - N + 2J \leq 1 + 28 - 3 + 3 = 29.$$

This contradicts $m \geq 30$. Therefore (2.6) proves $W_4 \geq 0$.

2.8 No full-support corner

Theorem 3.14, proved in Section 3.3, proves $\kappa_c \geq 1/3$ for every positive-volume finite-step central-box/horn set in the unit simplex. That section proves that degree at most four in this discrete class implies cardinality at most 29, by eleven explicit central-cap bounds.

If $D_0 < m - 1$ at $m \geq 30$, then $H \geq 5$, since $R \leq H < 5$ would force degree at most four. Sort weights to $(1, b, d)$, so $1 \leq b \leq d \leq H$. The continuous bound and (2.10) give the first row of Table 1, so Lemma 2.1 gives $H < 24$. For any allowance Q , the continuous theorem also gives $D_Q/m \geq (Q - 2B)/3$.

Finite lemma 2.3 (No-corner interval). For all real $1 \leq b \leq d \leq Q$, $5 \leq Q \leq 24$, and every nonempty finite no-full-corner ideal of attained height at most Q ,

$$m - D_Q \leq 1.$$

This is stronger than the genuine branch domain: no units, residue labels, attained allowance, or cardinality restriction is required. Use the method in Section 2.5, with no clipping. A DP leaf requires its exact bound at most q . An analytic leaf uses $Q_0 \geq 2(q + B_1 + C_1) + 3q$, which implies $D_Q \geq m$. The interval certificate covers the closed root from height 5 to 24. The finite lemma contradicts the hypothetical failure. Thus $D_0 \geq m - 1$ throughout this branch.

2.9 The short corner

Orient $p = (2, 1, 1)$, without prescribing the least weight's direction. For any degree allowance $r \geq 3$,

$$|T| \leq 2r^2 - r + 3. \quad (2.17)$$

Indeed the degree- r simplex minus the p -orthant has $2r^2 + 2$ points. For each $1 \leq j \leq r - 1$, the excluded point $(1, j, r - j)$ cannot dominate the unique full-support corner. It must dominate a planar corner; at least one of $(0, j, r - j)$, $(1, 0, r - j)$, $(1, j, 0)$ is therefore omitted. These $r - 1$ triples are pairwise disjoint, forcing (2.17). In particular $H < 6$ gives $R \leq 5$ and $30 \leq m \leq 48$. The endpoint $H = 6$ belongs to the high-height case.

2.9.1 High height

For a translated planar lower ideal of offset h in height H , the dimension-two line identity gives mean weight at most $(2H + h)/3$. Partition T into $y = 0$; $y \geq 1, z = 0$; $x = 0, y, z \geq 1$; and $x = 1, y, z \geq 1$. These are translated planar lower ideals, with offsets respectively 0, w_2 , $w_2 + w_3$, B . Thus $D_0/m \geq (H - 4B)/3$. If $H \geq 6$ and $D_0 < m - 1$, Lemma 2.1 with the second row of Table 1 gives $H < 42$. We retain the closed height-42 superset.

Finite lemma 2.4 (Short-corner high). For all $1 \leq b \leq d \leq Q$, $6 \leq Q \leq 42$, each coordinate permutation of $p = (2, 1, 1)$, and every nonempty finite no-full-corner U , the clipped $T = U \setminus (p + \mathbb{N}^3)$ of height at most Q satisfies $m - D_Q \leq 1$.

Apply Section 2.5's retained-height feasibility and clipped scores to all three corners. A DP leaf requires every corner bound at most q . The planar rule $Q_0 \geq 4(q + B_1 + C_1) + 3q$ is also sound, including equality. The complete closed tree establishes the finite lemma, contradicting failure.

2.9.2 Low height: exact profiles and two inline exceptions

A mixed xy corner has representative $\gamma e_3 \neq 0$. If its x -coordinate is at least two, subtracting $2e_1 + e_2$ gives an included point with residue $(\gamma + 1)a_3$. It forces $\gamma = h_3$; at most one such corner exists. The antichain permits at most one additional corner at $x = 1$. The xz plane is identical. A yz corner yields residue $(\gamma + 2)a_1$ on subtracting $e_2 + e_3$, forcing $\gamma \in \{h_1 - 1, h_1\}$. Hence each plane has at most two mixed corners, with the refined incident plane restriction, and contains respectively $(2, 1), (2, 1), (1, 1)$.

A degree-five plane profile is $f = (f_0, \dots, f_5)$, $0 \leq f_i \leq 6 - i$, nonincreasing, encoding $\{(i, j) : j < f_i\}$. Let $\ell(f)$ count positive columns and $\Delta(f) = \{i : 1 \leq i < \ell(f), f_i < f_{i-1}\}$. These, and only these, are mixed corners; a drop to zero is a pure-axis corner.

For xy, xz , require $f_2 \geq 2$, $|\Delta(f)| \leq 2$, and at most one drop at index at least two. For yz , require $h_1 \geq 2$ and $|\Delta(h)| \leq 2$. Use all compatible ordered triples f, g, h , with

$$\ell(f) = \ell(g), \quad f_0 = \ell(h), \quad g_0 = h_0,$$

and reconstruct

$$T = \{(x, y, z) : y < f_x, z < g_x, z < h_y, (x, y, z) \not\geq (2, 1, 1)\}. \quad (2.18)$$

Keep exactly those of degree at most five and $30 \leq |T| \leq 48$. Matching axes recover each plane section exactly. The required plane entries include all three predecessors of p . Any extra reconstructed point in a genuine ideal would dominate a planar minimal exclusion, violating its corresponding pair test. Thus every genuine low-height ideal is recovered uniquely. Profiles alone do not guarantee total degree: the literal degree check is essential.

Finite lemma 2.5 (Short-corner low). Every such triple satisfies $U_2 \geq m - 1$, except possibly the two keys

$$\begin{aligned} E_0 &= ((4, 3, 3, 2, 0, 0), (6, 3, 3, 2, 0, 0), (6, 4, 2, 2, 0, 0)), \\ E_1 &= ((6, 3, 3, 2, 0, 0), (4, 3, 3, 2, 0, 0), (4, 4, 2, 2, 1, 1)). \end{aligned}$$

For E_0 , direct reconstruction gives $m = 30$, $s = (27, 25, 34)$.

The included points $(0, 0, 5), (2, 2, 0), (3, 1, 0)$ with masses 38, 48, 4 have total mass 90 and residual $(108, 100, 190) - 4s = (0, 0, 54)$. Section 2.2 therefore gives $D_Q \geq 54 \geq 29$ for all admissible real weights. The second ideal and witness are obtained by interchanging y, z , including transposition of the yz profile.

Together with (2.4), the finite lemma proves $D_0 \geq m - 1$ in low height. This completes the short-corner branch.

2.10 Residual degree at most six

Let $5 \leq P = |p|_1 \leq 7$, $R \leq 6$. In an ij -plane, distinct mixed corners have distinct representatives $\gamma e_k \neq 0$. At most $p_i - 1$ have coordinate $q_i < p_i$, and at most $p_j - 1$ have $q_j < p_j$, by antichain distinct coordinates. For the remaining corners, subtract $p_i e_i + p_j e_j$. The included result has residue $(\gamma + p_k)a_k$; collision with the k -axis forces $\gamma > h_k - p_k$. At most p_k distinct representatives remain. Each plane thus has at most

$$(p_i - 1) + (p_j - 1) + p_k = P - 2 \quad (2.19)$$

mixed corners.

Define $F_i = \{x \in T : x + e_i \notin T\}$, $n_k = 1 + \max_T x_k$. For every residue-bijective lower ideal,

$$|F_i \cap F_j| \leq 2n_k. \quad (2.20)$$

To prove it, let C_{ij} contain excluded points with both positive i, j coordinates and both respective predecessors included. In each nonempty fixed- k slice, a planar ideal with h maxima has $h - 1$ mixed corners, by its constant positive-height blocks. All n_k slices are nonempty, hence $|C_{ij}| = |F_i \cap F_j| - n_k$. Subtracting an included predecessor forces the representative of each $q \in C_{ij}$ onto the k -axis, and distinct q have distinct representatives by comparing their i -predecessors. This injects C_{ij} into the n_k axis points, proving (2.20). Slice corners need not be globally minimal; only these two predecessors are used.

Sort coordinates of p , simultaneously permuting weights. The nine corners are

$$(1, 1, 3), (1, 2, 2), (1, 1, 4), (1, 2, 3), (2, 2, 2), (1, 1, 5), (1, 2, 4), (1, 3, 3), (2, 2, 3).$$

Use all nonincreasing seven-entry profiles $0 \leq f_i \leq 7 - i$. In an ij -plane require $f_{p_i} \geq p_j + 1$ and at most $P - 2$ positive height drops. Impose the three shared-axis compatibilities as in (2.18), and reconstruct with its own p -orthant deleted. Keep precisely degree at most six, cardinality at least 30, and the three bounds (2.20). No maxima, realizability, modular, or erosion filter is used. The same plane-recovery and excluded-corner argument as for (2.18) proves complete, unique coverage of genuine oriented ideals.

Finite lemma 2.6 (Degree-six). On this complete geometric family,

$$\max(U_2, G) \geq m. \quad (2.21)$$

The local integer predicate and the denominator-cleared axis predicate are exact. With $d = q_1 q_2 q_3 > 0$, the latter is

$$q_* \left(3md - 4 \sum_i s_i \prod_{j \neq i} q_j \right) \geq md.$$

Section 2.2 and (2.21) give $D_0 \geq m$ for this branch.

2.11 Higher degree: a finite strip and a real interval bound

Assume $R \geq 7$, $|p|_1 \geq 5$. We prove $D_0 \geq m - 29/10$. The first finite premise supplies a continuous inequality; that inequality restricts a hypothetical failure to a compact real parameter region. The second finite premise excludes the entire region.

2.11.1 The three-allowance strip

Finite lemma 2.7 (Strip). For $r \in \{18, 19, 20\}$, every finite lower ideal $T' \subseteq \mathbb{N}^3$ of degree at most r , with at most one positive minimal exclusion, satisfies

$$3r|T'| - 4 \sum_{x \in T'} |x|_1 \geq 4|T'|. \quad (2.22)$$

Empty ideals and missing coordinate units are allowed.

The empty ideal satisfies the inequality directly. Here is the exhaustive finite reduction for nonempty ideals. If there is a positive minimal corner p' , its predecessors imply $|p'|_1 \leq r+1$. Remove that exclusion generator to obtain a finite no-corner completion: pure-axis exclusions remain, and deleting a positive orthant leaves every axis point unchanged. Thus its axis caps are at most r . If there is no positive corner, take $p' = (1, 1, r-1)$; its orthant is inactive on the degree- r simplex. By symmetry enumerate exactly

$$1 \leq p'_1 \leq p'_2 \leq p'_3, \quad |p'|_1 \leq r+1.$$

No minimality filter is imposed on this larger family.

Apply the decomposition of Section 3.1, clipped by p' , to the score $f_r(x) = 4|x|_1 + 4 - 3r$. All center and transverse caps range from zero to r . A section or center is feasible when its **retained** maximum degree is at most r , not when its removed upper corner is. Apply Lemma 3.5 with the section sum of this score, retained-degree feasibility, and termination beyond the degree allowance. The maximum of center score plus its three horn bounds is an upper bound for $\sum_{T'} f_r = 4|T'| - D_r(T')$. Every configuration must have this bound at most zero. This is a search over corner/allowance configurations and exact dynamic-programming states. Singletons, zero caps, and inactive corners are included.

2.11.2 Transfer to the continuous gap

Let K be a positive-volume downset in the unit simplex whose complement is a finite union of upper orthants with at most one positive minimal vertex. Put $h = 1/21$, and for $z \in [0, h]^3$ define

$$T_z = \{y \in \mathbb{N}^3 : z + hy \in K\}, \quad r_z = \lfloor (1 - |z|_1)/h \rfloor, \quad \epsilon_z = 1 - |z|_1 - hr_z.$$

For $z \neq 0$, $0 < |z|_1 < 3h$, so $r_z \in \{18, 19, 20\}$ and $0 \leq \epsilon_z < h$. The zero translation is a null set and is omitted throughout the integral; no allowance-21 assertion is needed. Each excluded vertex g induces the discrete vertex

$$g'_i = \max(0, \lceil (g_i - z_i)/h \rceil).$$

A zero coordinate stays zero. Consequently at most one induced minimal vertex is positive, even after redundant vertices are removed. For almost every translation boundary conventions agree; the exceptional translations lie in finitely many coordinate hyperplanes modulo h . The simplex condition gives $|y|_1 \leq r_z$ for every $y \in T_z$. The hypotheses of (2.22) therefore hold for T_z , including when units are absent. Expansion gives

$$\sum_{y \in T_z} (3 - 4|z + hy|_1) = hD_{r_z}(T_z) + (3\epsilon_z - |z|_1)|T_z| \geq (4h - |z|_1)|T_z|.$$

Integrating the partition $x = z + hy$, whose Jacobian is one, gives

$$3 - 4\mathbb{E}_K|X|_1 \geq 4h - \sum_i \mathbb{E}_K(X_i \bmod h).$$

A coordinate fiber of a downset is an interval $[0, L]$ up to endpoints. Write $L = qh + t$, $0 \leq t < h$. Since

$$\int_0^L (x \bmod h) dx = (qh^2 + t^2)/2 \leq hL/2,$$

Fubini implies each remainder average is at most $h/2$. Hence

$$\boxed{\kappa_c(K) \geq 5/42.} \tag{2.23}$$

The averaged remainder estimate is essential; a pointwise estimate alone does not give this constant. The rectangular thickening from Section 2.4 preserves excluded supports and meets every hypothesis above.

2.11.3 Necessary region for a failure

Sort the normalized weights to $w = (1, b, d)$, and write $B = 1 + b + d$. The predecessors of the positive corner contain the units, so $1 \leq b \leq d \leq H$, and $H \geq R \geq 7$. If $D_0 < m - 29/10 < m$, Section 2.4 and (2.23) imply

$$B < 9 + \frac{28}{H-3}, \quad \frac{D_0}{m} \geq \frac{5H-37B}{42}, \quad 5H < 42 + 37B.$$

Lemma 2.1, with the final row of Table 1, gives $H < 78$. Also $p - e_1 \in T$ gives $w \cdot p \leq H + 1$. It suffices to cover the closed necessary failure region

$$\mathcal{F} = \{(b, d, Q) : 1 \leq b \leq d \leq Q, 7 \leq Q \leq 78, 1 + b + d \leq 9 + 28/(Q-3), 5Q \leq 42 + 37(1 + b + d)\}. \quad (2.24)$$

Finite lemma 2.8 (High-height). For every $(b, d, Q) \in \mathcal{F}$, every positive integer p with $|p|_1 \geq 5$ and $(1, b, d) \cdot p \leq Q + 1$, and every nonempty finite no-corner lower ideal U , put $T' = U \setminus (p + \mathbb{N}^3)$. If $\max_{T'}(1, b, d) \cdot x \leq Q$, then

$$|T'| - \left(3Q|T'| - 4(1, b, d) \cdot \sum_{T'} x \right) \leq 29/10. \quad (2.25)$$

No residue, cardinality, minimality, unit, or surface restriction is used in this enlarged geometric family. Use the following interval tightening. At scale $q = 4096$ write endpoints $L = (B_0, C_0, H_0)$, $U = (B_1, C_1, H_1)$ for (b, d, Q) . First intersect with order:

$$L \leftarrow (B_0, \max(B_0, C_0), \max(B_0, C_0, H_0)), \quad U \leftarrow (\min(B_1, C_1, H_1), \min(C_1, H_1), H_1).$$

If not inverted, compute, in this order,

$$\begin{aligned} S_{\max} &= 9q + \lceil 28q^2/(H_0 - 3q) \rceil, \\ B_1 &\leftarrow \min(B_1, \lceil (S_{\max} - q)/2 \rceil), \\ C_1 &\leftarrow \min(C_1, S_{\max} - q - B_0), \\ S_{\text{upper}} &= \min(q + B_1 + C_1, S_{\max}), \\ H_1 &\leftarrow \min(H_1, \lceil (42q + 37S_{\text{upper}})/5 \rceil). \end{aligned} \quad (2.26)$$

Perform at most twenty iterations, stopping on inversion or unchanged endpoints. Every upper rounding is outward. The inequalities in (2.24), the lower bound $qQ \geq H_0$, and $b \leq d$ prove that each step retains every relevant real point. Convergence is not required for soundness.

For a nonempty tightened box use Section 2.5 with $\ell = (q, B_0, C_0)$, $u = (q, B_1, C_1)$, and $\psi(x) = 4u \cdot x + q - 3H_0$. Enumerate the **full Cartesian** corner list

$$p_i \geq 1, \quad 5 \leq |p|_1 \leq \lfloor H_1/q \rfloor + 1, \quad \ell \cdot p \leq H_1 + q.$$

This contains every relevant corner, since $\ell_i \geq q$. An empty list is accepted only as vacuous. For every listed corner the retained-center/three-horn bound V_p must satisfy $10V_p \leq 29q$. The closed root is $((q, q, 7q), (78q, 78q, 78q))$. Each child is derived from its tightened parent; each split is strictly interior, and its two children share the split wall. Checked empty leaves and checked dynamic-programming leaves exhaust all nodes. Induction proves (2.25) throughout the closed real parameter region. At $Q = H$, it contradicts the proposed failure and proves this branch.

2.12 Assembly of the proof

Proof of Theorem 1.1. The branches are exhaustive. Multiplicity at most 29 is settled by Section 2.6. For $m \geq 30$, Section 2.7 routes every ideal to exactly one of Sections 2.8–2.11 or the analytic three-plane case. The three-plane case already gives $W_4 \geq 0$. On the remaining branches their conclusions give $D_0 \geq m - 29/10$. This bound is positive, and $A \geq m + 1$. Hence (2.1) gives

$$\begin{aligned} mW_4 &\geq (m+1) \left(m - \frac{29}{10} \right) - m(m-1) \\ &= -\frac{9m+29}{10} > -m, \end{aligned}$$

where the strict inequality is equivalent to $m > 29$. Thus $W_4 > -1$, and its integrality gives $W_4 \geq 0$. More generally, the same calculation would accept a uniform deficit $D_0 \geq m - C$ for all $m \geq 30$ whenever $C < 90/31$; the retained value $C = 29/10$ lies just below this threshold. Together with the direct analytic three-plane case and the small-multiplicity result, this proves the theorem using the stated, independently replayed finite lemmas and the published multiplicity-at-most-19 result. $\square$

3 Analytic details

The following three arguments are integral to the proof.

3.1 Central boxes and three monotone horns

3.1.1 Statement

For $r \in \mathbb{N}$, write $[r] = \{0, 1, \dots, r\}$. For $z \in \mathbb{N}^3$, let

$$\mathcal{B}(z) = [z_1] \times [z_2] \times [z_3].$$

Theorem 3.1 (Central-box/three-horn decomposition). *Let $U \subseteq \mathbb{N}^3$ be a nonempty finite lower ideal. Suppose that no minimal point of $\mathbb{N}^3 \setminus U$ has all three coordinates positive. Then there is a point $c \in U$ and a disjoint partition*

$$U = \mathcal{B}(c) \dot{\cup} A_1 \dot{\cup} A_2 \dot{\cup} A_3 \tag{3.1}$$

with the following properties. For each i , let j, k be the other two coordinates, and put $h_i = \max\{x_i : x \in U\}$, which is also the largest coordinate- i axis level in U . Then $h_i \geq c_i$, and there are nonincreasing functions

$$r_i : \{c_i + 1, \dots, h_i\} \longrightarrow \{0, \dots, c_j\}, \quad s_i : \{c_i + 1, \dots, h_i\} \longrightarrow \{0, \dots, c_k\} \tag{3.2}$$

such that

$$A_i = \left\{ x \in \mathbb{N}^3 : c_i < x_i \leq h_i, \quad 0 \leq x_j \leq r_i(x_i), \quad 0 \leq x_k \leq s_i(x_i) \right\}. \tag{3.3}$$

The domain in (3.2) is empty when $h_i = c_i$, giving an empty horn. Thus the theorem includes a single box, one or two nonempty horns, zero central caps, and ideals that do not contain all coordinate units.

3.1.2 Plane determination and the compatibility graph

The exclusion hypothesis has the following equivalent form:

$$x \in U \iff (x_1, x_2, 0), (x_1, 0, x_3), (0, x_2, x_3) \in U. \quad (3.4)$$

The forward implication is lower closure. Conversely, if $x \notin U$, then x dominates a minimal excluded point p . At least one coordinate of p is zero, so one of the three planar points in (3.4) also dominates p and lies outside U .

For coordinate i , let

$$h_i^{\text{ax}} = \max\{t \in \mathbb{N} : te_i \in U\}.$$

Construct a graph G with vertices

$$V_i = \{(i, t) : 0 \leq t \leq h_i^{\text{ax}}\}, \quad i = 1, 2, 3.$$

Each V_i is a clique. For $i \neq j$, join (i, t) to (j, s) exactly when $te_i + se_j \in U$. The three zero-level vertices are universal. Moreover, lower closure gives the nested-neighborhood property

$$t' < t \implies N(i, t) \setminus V_i \subseteq N(i, t') \setminus V_i. \quad (3.5)$$

Lemma 3.2. *The graph G is chordal.*

Proof. Suppose an induced cycle has length at least four. Because there are only three coordinate classes, it repeats a class. An induced cycle can contain at most two vertices from a clique, and those two must be consecutive. Write their levels as $t' < t$. The other cycle neighbor of (i, t) lies outside V_i ; by (3.5), it is also adjacent to (i, t') . This is a chord, a contradiction. $\square$

3.1.3 The required clique-tree lemma

Lemma 3.3 (Clique tree). *The distinct maximal cliques of a finite nonempty connected chordal graph are the nodes of a tree such that the nodes containing any fixed graph vertex form a connected subtree.*

Proof. We induct on the number of graph vertices. A complete graph has a one-node tree. Otherwise choose nonadjacent vertices a, b and an inclusion-minimal set Q separating them. Let C_a, C_b be their components in $G - Q$.

The separator Q is a clique. Indeed, minimality implies that every $q \in Q$ has a neighbor in both C_a and C_b . If nonadjacent $q, r \in Q$ existed, take a shortest q -to- r path whose internal vertices lie in C_a , and another through C_b . Their union would be an induced cycle of length at least four: shortestness rules out chords within either path, and distinct components of $G - Q$ have no edges between them. This contradicts chordality.

Let $C_1, \dots, C_d$, with $d \geq 2$, be all components of $G - Q$. Each induced graph on $C_\ell \cup Q$ is connected, chordal, and smaller. Apply induction to each, and choose in each resulting clique tree a bag K_ℓ containing the clique Q . Join the nodes K_ℓ in a star. A vertex outside Q occurs in only one component tree. A vertex in Q occurs in a connected subtree of every component tree containing K_ℓ , and the added star joins those subtrees. Thus the resulting tree of clique bags has the required connectedness property.

Every maximal clique of G occurs among these bags, because a clique cannot meet two components of $G - Q$. Some bags may be nonmaximal in G , or the same maximal clique may occur more than

once. If a bag B is contained in another bag C , then every node on the tree path from B to C contains every vertex of B . Contract the first edge of that path and keep the neighboring bag. This preserves the connectedness property. Repeating removes all nonmaximal and duplicate bags, leaving exactly the distinct maximal cliques. $\square$

This finite clique-tree property is classical; see Gavril [Gav74].

3.1.4 From maximal cliques to maximal boxes

Every maximal clique of the compatibility graph contains a prefix of each coordinate class. Indeed, if it contains (i, t) , then every lower level in V_i is adjacent to the whole clique by (3.5), and maximality adds it. The universal zero levels ensure that no coordinate class is absent.

Let z_i be the largest level from V_i in the clique. All three pairs of coordinate maxima are adjacent, so (3.4) gives $z = (z_1, z_2, z_3) \in U$. Conversely, every $z \in U$ gives the clique consisting of the three coordinate prefixes through z_i . Inclusion of these prefix cliques is exactly coordinatewise comparison of their upper corners. It follows that the maximal cliques of G correspond bijectively to the maximal points of U , and hence to the anchored boxes $\mathcal{B}(z)$ maximal in U : a nonmaximal corner extends its prefix clique, and any extension of a prefix clique produces a larger corner through (3.4).

Apply Lemma 3.3, and write $\mathcal{T}$ for the resulting tree on maximal points. A graph vertex (i, t) belongs to the clique for z precisely when $z_i \geq t$. Therefore

$$\boxed{\{z \in \text{Max}(U) : z_i \geq t\} \text{ induces a connected subtree whenever it is nonempty.}} \quad (3.6)$$

3.1.5 The median and the three arms

Distinct maximal points of U are incomparable. Let $z(v)$ denote the point labeling a node v of $\mathcal{T}$.

If v is a leaf with neighbor w , choose a coordinate i for which $z_i(v) > z_i(w)$. If another node u had $z_i(u) \geq z_i(v)$, the superlevel subtree (3.6) at level $z_i(v)$ would contain u, v but not w , even though the path from u to v passes through w . Hence v is the unique global maximizer of coordinate i . Two leaves cannot be unique maximizers of the same coordinate, so $\mathcal{T}$ has at most three leaves.

For each coordinate i , choose a node v_i maximizing z_i . Every leaf is one of v_1, v_2, v_3 , by the preceding uniqueness. Consequently the whole tree is the minimal subtree spanning those three nodes: otherwise a component outside that spanning subtree would end in an unselected leaf.

Let v_0 be the median of v_1, v_2, v_3 , the unique common node of their three pairwise paths, and set $c = z(v_0)$. Repeated selected nodes are allowed. The nontrivial paths from v_0 to the v_i cover the tree and meet only at v_0 ; these are the at most three arms.

Consider an edge uv on the path from v_0 toward v_i , oriented away from v_0 . If $z_i(u) > z_i(v)$, the coordinate- i superlevel subtree at height $z_i(u)$ would contain u, v_i but not the intervening node v . Thus

$$z_i(u) \leq z_i(v). \quad (3.7)$$

For $j \neq i$, the selected maximizer v_j is either v_0 or lies on another arm. Hence u lies on the path from v to v_j . If $z_j(v) > z_j(u)$, the corresponding superlevel subtree would contain v, v_j but not u . Therefore

$$z_j(u) \geq z_j(v), \quad j \neq i. \quad (3.8)$$

Equality in (3.7), together with (3.8), would give $z(v) \leq z(u)$, contrary to incomparability. Thus the outward coordinate increases strictly along its arm, while both transverse coordinates weakly decrease.

3.1.6 Disjoint rectangular sections

All maximal boxes on arm i have transverse caps at most c_j, c_k . Fix an integer $c_i < t \leq h_i$. The central node and the other arms have coordinate- i cap at most c_i , so they contribute nothing at this level. Among nodes on arm i whose outward cap is at least t , let $v(t)$ be the first. All later contributing boxes have transverse rectangles contained in that of $v(t)$, by (3.8). Hence the entire level section is

$$\{t\} \times \prod_{\ell \neq i} [z_\ell(v(t))], \quad (3.9)$$

after putting coordinate i first. As t increases, $v(t)$ moves weakly outward, so both transverse caps are nonincreasing. There is no gap between $c_i + 1$ and the last nonempty level, because every anchored box contains all lower levels.

Every point of U belongs to a maximal anchored box. If it is outside $\mathcal{B}(c)$, that maximal box lies on a unique arm, and the point exceeds the corresponding central cap while staying below the other two. It therefore lies in exactly one section (3.9). This proves the disjoint partition (3.1) and Theorem 3.1. $\square$

Positive diagonal scaling preserves the decomposition. Rectangular thickening replaces each integer level by a slab and each integer transverse rectangle by a continuous rectangle; its two transverse side lengths remain nonincreasing. Thus the thickened sets of Section 2.4 belong to the same central-box/three-horn geometric class, up to null boundary conventions.

3.1.7 One full-support corner

Let $T \subseteq \mathbb{N}^3$ be a finite lower ideal with exactly one full-support minimal excluded point p . Let P be the set of all minimal excluded points of T , and define

$$U = \mathbb{N}^3 \setminus \bigcup_{q \in P \setminus \{p\}} (q + \mathbb{N}^3). \quad (3.10)$$

Corollary 3.4 (One-corner clipping). *The set U is a finite lower ideal with no full-support minimal exclusion, and*

$$\boxed{T = U \setminus (p + \mathbb{N}^3)}. \quad (3.11)$$

The ideals T and U have the same points on every coordinate axis. If (3.1) is a central-box/three-horn partition of U , then

$$T = (\mathcal{B}(c) \setminus (p + \mathbb{N}^3)) \dot{\cup} (A_1 \setminus (p + \mathbb{N}^3)) \dot{\cup} (A_2 \setminus (p + \mathbb{N}^3)) \dot{\cup} (A_3 \setminus (p + \mathbb{N}^3)). \quad (3.12)$$

At an outward level t of horn i , with transverse coordinates j, k and uncut caps r, s , the retained section is the exact set

$$\begin{cases} [r] \times [s], & t < p_i, \\ ([r] \times [s]) \setminus ([p_j, r] \times [p_k, s]), & t \geq p_i, \end{cases} \quad (3.13)$$

where an interval $[a, r]$ is empty when $a > r$.

Proof. Every member of $P \setminus \{p\}$ has support at most two, so the minimal excluded points of U , after redundant generators are removed, also have support at most two. For each coordinate i , the first missing axis point of T is a pure-axis member of P . It remains in (3.10), proving that U is finite. The full-support orthant $p + \mathbb{N}^3$ contains no axis point, so T and U have the same axis sections. The upper-ideal description of a lower ideal's complement gives (3.11).

Deleting the same upper orthant from each piece of a disjoint partition preserves disjointness and proves (3.12). At level $t < p_i$, no point lies above p ; at level $t \geq p_i$, precisely the transverse upper-right rectangle lies above p . This proves (3.13). $\square$

The same clipping formulas remain valid for any positive integer point p , even if it is inactive or not minimal excluded after clipping. Allowing such choices enlarges a later optimization class and therefore remains safe for an upper-bound certificate. In particular, if no point of U dominates p , the clipping is inactive and the representation includes the no-full-corner case.

3.1.8 The nested-horn recurrence

Lemma 3.5 (Nested-horn recurrence). *Fix an outward coordinate i , a first level $t_0 = c_i + 1$, and central transverse caps $b_0 = c_j$, $d_0 = c_k$. Suppose a problem assigns an additive score $C_i(t, r, s)$ to an uncut rectangular section or the retained section (3.13), and has a pointwise feasibility predicate $\mathcal{F}_i(t, r, s)$. Let $F_i(t, b, d)$ be the maximum score of a possibly empty nested sequence starting at level t , with first caps at most b, d . Then*

$$F_i(t, b, d) = \max \left\{ 0, \max_{\substack{0 \leq r \leq b, \\ 0 \leq s \leq d, \\ \mathcal{F}_i(t, r, s)}} (C_i(t, r, s) + F_i(t + 1, r, s)) \right\}. \quad (3.14)$$

A terminal level beyond the allowed axis range has value zero.

Proof. The terminal value is the induction base case. The zero option terminates the horn. Every nonempty uncut section contains its axis point, which survives clipping because p is positive; lower closure then forbids a later nonempty section after termination. Every other choice in (3.14) selects the current rectangle and passes exactly its caps to the next level. Induction on the remaining number of levels proves that (3.14) covers every feasible nested horn and only such sequences. There are finitely many levels and cap choices, so backward evaluation terminates. $\square$

Equivalently, a prefix-maximum implementation may compare the state (b, d) with $(b - 1, d)$ and $(b, d - 1)$, because every proper subrectangle decreases at least one cap.

For a fixed center, the three arms in (3.1) or (3.12) are disjoint, so their additive optima may be computed independently and added to the central-piece score. Maximizing over all centers therefore covers every ideal in the theorem or clipping corollary. Particular score formulas, degree or weighted height allowances, and interval bounds are specified in Section 2.

3.2 Full weighted Apéry ideals

3.2.1 Setup and statement

Let

$$S = \langle m, a_1, a_2, a_3 \rangle$$

be a minimally four-generated numerical semigroup. Let $T \subseteq \mathbb{N}^3$ be its preferred Apéry lower ideal, and put

$$\lambda(x) = a_1x_1 + a_2x_2 + a_3x_3, \quad M = \max_{x \in T} \lambda(x).$$

Thus $|T| = m$, the labels of its points are pairwise distinct modulo m , and $0, e_1, e_2, e_3 \in T$. We say that T is *full weighted* if

$$T = \{x \in \mathbb{N}^3 : \lambda(x) \leq M\}. \quad (3.15)$$

Permuting the three coordinates preserves (3.15), lower closure, residue injectivity, and all projection quantities used below. We may therefore write the three nonmultiplicity generators as

$$a < b < d$$

and use coordinates (x, y, z) , so that

$$\lambda(x, y, z) = ax + by + dz.$$

Theorem 3.6 (Full-weighted-ideal theorem). *If the preferred Apéry ideal T is full weighted, then*

$$W_4(S) \geq 0.$$

3.2.2 Classification of the column projection

Project away the x -coordinate. By (3.15), its image is

$$Q = \pi_x(T) = \{(y, z) \in \mathbb{N}^2 : by + dz \leq M\}. \quad (3.16)$$

Put

$$h_y = \left\lfloor \frac{M}{b} \right\rfloor, \quad h_z = \left\lfloor \frac{M}{d} \right\rfloor.$$

Both are positive because $e_2, e_3 \in T$.

Lemma 3.7 (Three possible projections). *The set Q is one of*

$$\begin{aligned} Q_3 &= \{(0, 0), (1, 0), (0, 1)\}, \\ Q_4 &= \{(0, 0), (1, 0), (2, 0), (0, 1)\}, \\ Q_6 &= \{(y, z) \in \mathbb{N}^2 : y + z \leq 2\}. \end{aligned} \quad (3.17)$$

Proof. For $1 \leq j \leq h_y$, define

$$p_j = \left(\left\lfloor \frac{M - bj}{a} \right\rfloor + 1, j, 0 \right). \quad (3.18)$$

This is a minimal excluded point. Its x -predecessor is included by the floor definition. Moreover,

$$\lambda(p_j) \leq M + a,$$

so its y -predecessor has weight at most $M + a - b \leq M$. Both displayed coordinates of p_j are positive because $bj \leq M$.

The representative in T of the residue of p_j must have support disjoint from $\{x, y\}$, so it lies on the z -axis. The corners $p_1, \dots, p_{h_y}$ have intersecting supports and hence distinct residues. The z -axis contains exactly

$$0, e_3, \dots, h_z e_3,$$

and therefore

$$h_y \leq h_z + 1. \quad (3.19)$$

First suppose $M < b + d$. Then (3.16) contains no point with both coordinates positive. Since

$$d \leq M < b + d < 2d,$$

we have $h_z = 1$. Equation (3.19) gives $h_y \leq 2$, while $h_y \geq 1$. This gives Q_3 or Q_4 .

Now suppose $M \geq b + d$. The point

$$p = \left(\left\lfloor \frac{M - b - d}{a} \right\rfloor + 1, 1, 1 \right) \quad (3.20)$$

is a full-support minimal exclusion: its x -predecessor is included by the floor definition, while removing e_2 or e_3 reduces its weight from at most $M + a$ by at least a . By the corner rules, p has residue zero. Each p_j has support intersecting that of p , so none of their distinct representatives on the z -axis can be zero. Hence

$$h_y \leq h_z. \quad (3.21)$$

If $M \geq 2b + d$, then

$$p' = \left(\left\lfloor \frac{M - 2b - d}{a} \right\rfloor + 1, 2, 1 \right)$$

is a second full-support minimal exclusion. Indeed, its x -predecessor is included by the floor definition, and removing e_2 or e_3 reduces its weight from at most $M + a$ by at least a . It is distinct from p because its y -coordinate is two. This contradicts the uniqueness of a full-support corner, so

$$M < 2b + d < 3d$$

and $h_z \leq 2$. On the other hand,

$$M \geq b + d > 2b,$$

so $h_y \geq 2$. Since $b < d$ also gives $h_y \geq h_z$, equation (3.21) forces

$$h_y = h_z = 2. \quad (3.22)$$

It follows from (3.22) that $2d \leq M < 3b$. Every (y, z) of total degree at most two has weight at most $2d$, while every point of total degree at least three has weight at least $3b$. Thus (3.16) is exactly Q_6 . □

3.2.3 The six-column triangle is impossible

For $(y, z) \in Q$, let

$$L_{yz} = \left\lfloor \frac{M - by - dz}{a} \right\rfloor + 1 \quad (3.23)$$

be the length of its x -column. Subtracting b or d from the available weight decreases a positive column length by at least one, because $b, d > a$.

Lemma 3.8 (Six-column exclusion). *The projection Q_6 in (3.17) cannot occur.*

Proof. Suppose it does, and abbreviate its positive column lengths by

$$\begin{array}{c|cccccc} (y, z) & (0, 0) & (1, 0) & (2, 0) & (0, 1) & (1, 1) & (0, 2) \\ \hline L_{yz} & \alpha & \beta & \gamma & \varphi & \varepsilon & \eta. \end{array} \quad (3.24)$$

The strict column descents give

$$\alpha > \beta > \gamma > 0, \quad \alpha > \varphi > \eta > 0, \quad \varepsilon < \min(\beta, \varphi). \quad (3.25)$$

Work in $\mathbb{Z}/m\mathbb{Z}$, and let r be the least positive integer for which $ra \equiv 0 \pmod{m}$. The point $(\varepsilon, 1, 1)$ is the full-support minimal exclusion above the mixed column: its x -predecessor lies in that column, and its other two predecessors are included by $\varepsilon < \varphi, \beta$. It therefore has residue zero:

$$\varepsilon a + b + d \equiv 0 \pmod{m}. \quad (3.26)$$

The residues in the origin column are

$$0, a, \dots, (\alpha - 1)a,$$

whereas (3.26) makes those in the mixed column

$$-\varepsilon a, -(\varepsilon - 1)a, \dots, -a.$$

All $\alpha + \varepsilon$ points belong to T , so residue injectivity shows that these consecutive multiples of a are distinct. Consequently

$$r \geq \alpha + \varepsilon. \quad (3.27)$$

The two points $(\beta, 1, 0)$ and $(\gamma, 2, 0)$ are minimal exclusions: their x -predecessors lie at the tops of their columns, and their y -predecessors are included by $\beta < \alpha$ and $\gamma < \beta$. Their representatives have support on the z -axis. Their residues are nonzero because their supports meet that of $(\varepsilon, 1, 1)$, and they are distinct from each other. Since the z -axis consists of $0, e_3, 2e_3$, their residues therefore permute $d, 2d$.

The swapped assignment would be

$$\beta a + b \equiv 2d, \quad \gamma a + 2b \equiv d. \quad (3.28)$$

Subtracting twice the second congruence in (3.28) from the first gives

$$(\beta - 2\gamma)a - 3b \equiv 0 \pmod{m},$$

while the second congruence in (3.28), substituted into (3.26), gives

$$(\varepsilon + \gamma)a + 3b \equiv 0 \pmod{m}.$$

Adding these relations eliminates b and gives

$$(\beta - \gamma + \varepsilon)a \equiv 0 \pmod{m}.$$

But

$$0 < \beta - \gamma + \varepsilon < \alpha + \varepsilon \leq r,$$

contradicting the definition of r . We must therefore have

$$\beta a + b \equiv d, \quad \gamma a + 2b \equiv 2d. \tag{3.29}$$

The same argument for the minimal exclusions $(\varphi, 0, 1)$ and $(\eta, 0, 2)$, whose representatives lie on the y -axis, rules out the swapped assignment there: it would give $(\varphi - \eta + \varepsilon)a \equiv 0 \pmod{m}$, although

$$0 < \varphi - \eta + \varepsilon < \alpha + \varepsilon \leq r.$$

Consequently

$$\varphi a + d \equiv b, \quad \eta a + 2d \equiv 2b. \tag{3.30}$$

Adding the first congruences in (3.29)–(3.30), and subtracting the second congruences from twice their respective first congruences, shows that

$$\beta + \varphi, \quad 2\beta - \gamma, \quad 2\varphi - \eta \tag{3.31}$$

are positive multiples of r . Each is strictly less than 2α , and (3.27) gives $\alpha < r$. Thus all three integers in (3.31) lie strictly between zero and $2r$, so

$$\beta + \varphi = r, \quad 2\beta - \gamma = r, \quad 2\varphi - \eta = r.$$

Adding the last two equalities and subtracting twice the first gives

$$\gamma + \eta = 0,$$

contrary to (3.25). Hence Q_6 is impossible. $\square$

Remark 3.9. The proof of Lemma 3.8 used only residue bijectivity, the corner rules, the six-point projection, and the strict descents in (3.25). The exact floor formula (3.23) is not otherwise needed. This isolates the arithmetic obstruction from the full-weighted classification which produces the six-point shape.

3.2.4 The remaining projection scores

By Section 2.3, it suffices to prove $\Phi(T, K) \geq 0$, where $K = \text{Max}(T)$. The two remaining projections admit one calculation.

Lemma 3.10 (Three- and four-column projections). *If $Q = Q_k$, $k \in \{3, 4\}$, then $\Phi(T, K) \geq 0$.*

Proof. Let α be the origin-column length, and index the other lengths in the order displayed in (3.17). Since $b, d > a$, the lengths decrease strictly along each axis of the projection:

Projection	Column lengths	Strict descents
Q_3	α, β, γ	$\alpha > \beta, \gamma \geq 1$
Q_4	$\alpha, \beta, \gamma, \delta$	$\alpha > \beta > \gamma \geq 1, \alpha > \delta \geq 1$

These descents imply $\alpha \geq k - 1$ and make the column tops pairwise incomparable, so they are precisely K .

Every non-origin column lies on exactly one axis of the projection. The two projections retaining the column direction therefore count each column once and the origin column twice. Hence

$$|K| = k, \quad P(T) = k + \alpha + m. \quad (3.32)$$

For these two shapes the sum of the projected total degrees is $2k - 4$, and the sum of their coordinate maxima is $k - 1$. Thus

$$\sum_{x \in K} |x|_1 = m + k - 4, \quad \sum_j \max_{x \in K} x_j = \alpha + k - 2.$$

It follows that

$$E(K) = m - \alpha - 2, \quad \Phi(T, K) = 2(\alpha - k + 1) \geq 0. \quad (3.33)$$

□

3.2.5 Proof of the full-weighted theorem

Proof of Theorem 3.6. Lemma 3.7 lists the three possible x -column projections. Lemma 3.8 excludes the six-column triangle, and Lemma 3.10 gives $\Phi(T, K) \geq 0$ in the remaining cases. The projection inequality (2.6) proves $W_4(S) \geq 0$.

□

3.3 The continuous no-corner moment inequality

3.3.1 Box-horn sets and their moment functional

Boundary faces never affect the integrals below. A *box-horn set of height one* is the disjoint union, up to null sets, of a central box

$$C = [0, a] \times [0, b] \times [0, c], \quad a, b, c \geq 0, \quad a + b + c \leq 1, \quad (3.34)$$

and at most three outward horns. The horn beyond the a -face has the form

$$\{(t, y, z) : a < t < 1, 0 < y < f(t), 0 < z < g(t)\}, \quad (3.35)$$

where f, g are nonnegative, nonincreasing finite step functions satisfying

$$f(t) \leq b, \quad g(t) \leq c, \quad t + f(t) + g(t) \leq 1. \quad (3.36)$$

The other two horns are defined by coordinate permutation. Empty horns and zero-width sections are allowed.

For a measurable $E \subseteq \mathbb{R}_{\geq 0}^3$, define

$$\mathcal{J}(E) = \int_E (3(x_1 + x_2 + x_3) - 2) dx. \quad (3.37)$$

Thus $\mathcal{J}(E) \leq 0$ is equivalent to

$$\frac{1}{\text{vol } E} \int_E (x_1 + x_2 + x_3) dx \leq \frac{2}{3} \quad (3.38)$$

when E has positive volume. The central box contributes

$$\mathcal{J}(C) = abc \left(\frac{3}{2}(a + b + c) - 2 \right). \quad (3.39)$$

The contribution of the horn (3.35) is

$$\mathcal{J}_a(f, g) = \int_a^1 f(t)g(t) \left(3t + \frac{3}{2}(f(t) + g(t)) - 2 \right) dt. \quad (3.40)$$

We next obtain a sharp enough common envelope for (3.40), then show that the three envelopes are absorbed by the negative contribution (3.39).

3.3.2 The boundary-tail potential

It is convenient to give the formulas at an arbitrary height allowance $L > 0$. At this height write

$$\mathcal{J}_L(E) = \int_E (3(x_1 + x_2 + x_3) - 2L) dx,$$

so that $\mathcal{J} = \mathcal{J}_1$. For $0 \leq x \leq y$, define

$$\begin{aligned} P_L(x, y) &= \frac{Lx^3}{6} - \frac{x^4}{4} + \frac{Lxy^2}{2} - \frac{3x^2y^2}{4} - \frac{xy^3}{2}, \\ g(q, z) &= \frac{q^2z^2}{8} + \frac{q^3z}{24} + \frac{q^4}{192}, \\ Q_L(x, y) &= P_L(x, y) + g((y + 2x - L)_+, y - x). \end{aligned} \quad (3.41)$$

Extend Q_L symmetrically in x, y .

Lemma 3.11 (Boundary-tail optimum). *Suppose a horn is already on the height boundary: in reverse time $r = L - t$, its transverse sides are nondecreasing functions with sum r , ending at caps $x \leq y$. Its contribution is at most $Q_L(x, y)$.*

Proof. By scaling, take $L = 1$. At each r , sort the two sides and write their difference as $d(r) \geq 0$. Sorting preserves the sum, product, caps, and monotonicity. The boundary contribution becomes

$$\int_0^{x+y} \frac{(2 - 3r)(r^2 - d(r)^2)}{8} dr. \quad (3.42)$$

The two sides are nondecreasing precisely when d is 1-Lipschitz. The cap conditions give

$$\max(0, r - 2x) \leq d(r) \leq \min(r, 2y - r). \quad (3.43)$$

The coefficient of $-d^2$ changes sign only at $r = 2/3$. After fixing $d(2/3)$, the optimal admissible function is therefore the smallest one allowed by (3.43) to the left of $2/3$, and the largest one allowed to its right. Writing $q = (2/3 - d(2/3))/2$, the associated transverse sides are

$$\begin{cases} (r/2, r/2), & 0 \leq r \leq 2q, \\ (q, r - q), & 2q \leq r \leq y + q, \\ (r - y, y), & y + q \leq r \leq x + y. \end{cases} \quad (3.44)$$

The same profile follows directly from pointwise product maximization if $x + y \leq 2/3$. When $x + y \geq 2/3$, feasibility gives

$$\max(0, 2/3 - y) \leq q \leq \min(x, 1/3).$$

Let $I(q; x, y)$ denote the integral in (3.42) for (3.44). Direct integration gives

$$\frac{\partial I}{\partial q} = \frac{(y - q)^2(1 - y - 2q)}{2}. \quad (3.45)$$

Hence the optimum is attained at

$$q_* = \min\left(x, \frac{1 - y}{2}\right). \quad (3.46)$$

In the second case above, $y \geq 1/3$, so q_* lies in the displayed feasible interval. The endpoint $q = x$ has value $P_1(x, y)$. If $q_* < x$, integration of (3.45) from q_* to x shows that the improvement is

$$g(y + 2x - 1, y - x). \quad (3.47)$$

Equations (3.41) and (3.47) prove the claim for $L = 1$; homogeneity gives the general statement. $\square$

3.3.3 Reducing an arbitrary horn to one slab

For $a + x + y \leq L$, put

$$F_L(a; x, y) = Q_L(x, y) + \frac{3a - L}{2}xy(L - a - x - y), \quad (3.48)$$

and, whenever $a + B + C \leq L$, define

$$U_L(a; B, C) = \max_{0 \leq x \leq B, 0 \leq y \leq C} F_L(a; x, y). \quad (3.49)$$

The term added to Q_L is the contribution of a constant initial slab from $t = a$ to $t = L - x - y$.

Lemma 3.12 (One-slab envelope). *Every finite-step horn based at a , with transverse caps B, C and height allowance L , has contribution at most $U_L(a; B, C)$.*

Proof. Scale to $L = 1$, and sort the two transverse coordinates so $B \leq C$. Replacing the two sides pointwise by their minimum and maximum preserves nestedness and area. The smaller side remains at most B , while both original sides and hence the larger remain at most C . We may therefore assume that the first side is the smaller one at every level. We proceed in three stages: make the

slabs height-tight, compare them with a boundary interpolation, and remove or merge the remaining jumps.

Step 1: endpoint tightness. First fix the finite list of nested rectangle states and vary only the outward endpoints of their slabs. The objective is a separable convex quadratic in those endpoints: the quadratic coefficient at an endpoint is $3(A_i - A_{i+1})/2 \geq 0$, where A_i is the area of the i -th rectangle and $A_{n+1} = 0$. A maximum on the endpoint polytope is attained at a vertex. A block of equal endpoints at such a vertex either begins at a or meets one of its height bounds. Removing zero-length slabs leaves every surviving slab tight at its right endpoint.

Step 2: boundary interpolation. Reverse time by $r = 1 - t$, and write

$$A(r) = 1 - \frac{3}{2}r.$$

If a tight slab starts at reverse time r , ends with rectangle (x, y) , and $x + y = s \leq r$, its exact contribution is

$$A(r)(r - s)xy. \tag{3.50}$$

Retain the possibly slack initial slab, including any central slack; the filling below applies only to transitions between genuine tight states. For transitions wholly inside $r \leq 2/3$, any monotone boundary interpolation has area at least that of the smaller endpoint rectangle, while $A(r) \geq 0$. Filling those transitions therefore weakly increases the objective.

If none of the states reaches reverse time at most $2/3$, all terms in (3.50) are nonpositive and the empty horn dominates. Otherwise let (u, v) , $u \leq v$, be the first tight state with $u + v \leq 2/3$. From (u, v) toward reverse time zero, use the pointwise maximizing continuation: keep the smaller side u fixed down to (u, u) , then follow the balanced path. This continuation is nested and respects both caps; it is the boundary profile from Lemma 3.11 in this nonnegative region.

It remains to control the earlier transitions between genuine states, where $A(r) \leq 0$. At every such state of sum s , nestedness and the cap conditions allow its product to be replaced by the smaller product at

$$x_*(s) = \max(u, s - C), \quad y_*(s) = \min(C, s - u). \tag{3.51}$$

Indeed $x(s - x)$ is increasing for $x \leq s/2$; the pair in (3.51) is nested as s varies, respects B, C , and ends at (u, v) . Since the coefficient in (3.50) is nonpositive, this replacement can only increase the objective. All earlier states now lie on the L-shaped boundary path

$$(B, C) \longrightarrow (u, C) \longrightarrow (u, u). \tag{3.52}$$

The possibly slack initial position is represented only for bookkeeping by the virtual state (x_0, C) , where $x_0 = 1 - a - C \geq B$. The first actual jump covers the virtual excess.

Step 3: removing and merging jumps. We now compare a jump on (3.52) with filling the same part of the boundary path continuously. Along a branch with one fixed side c , a jump of length d has gain

$$G_c(d) = \frac{c}{4}d^2(2d + 3c - 2), \tag{3.53}$$

and two jumps satisfy

$$G_c(d + e) - G_c(d) - G_c(e) = \frac{3c}{2}de \left(d + e + c - \frac{2}{3} \right). \tag{3.54}$$

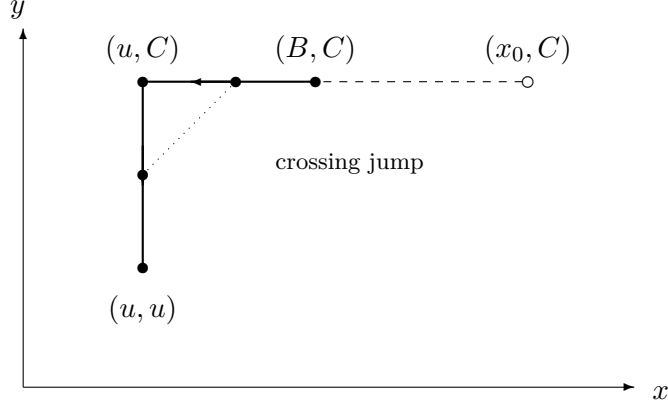


Figure 1: Schematic of the L-shaped boundary path in (3.52). The dashed extension represents the virtual initial state; a crossing jump skips the turn.

These follow by integrating $c(r - c)A(r)$.

Jumps after the turn. On the second branch of (3.52), the fixed side is u , and every jump has $d \leq C - u \leq 1 - 2u$. Thus $2d + 3u - 2 \leq -u$, so (3.53) is nonpositive and all such jumps may be filled.

A jump crossing the turn. At most one jump crosses the turn in (3.52), as illustrated in Figure 1. Suppose it starts at $(u + p, C)$ and ends at $(u, C - q)$. If its gain over the continuous L-path is denoted by $G(p, q; u, C)$, direct integration gives

$$\frac{\partial G}{\partial q} = \frac{u}{2}(p + q)(3p + 3q + 3u - 2). \quad (3.55)$$

This derivative changes sign at most once, from negative to positive, so the maximum occurs at $q = 0$ or $q = C - u$. The continuous baseline from the jump's start through the remaining tail is independent of q , so the same endpoint conclusion applies to the whole continuation. The first choice moves the jump entirely onto the fixed- C branch.

For the balanced endpoint, a jump from reverse time r to (z, z) , followed by the balanced tail, has value

$$E_r(z) = A(r)(r - 2z)z^2 + \frac{2}{3}z^3 - \frac{3}{2}z^4,$$

with

$$E'_r(z) = -6z \left(z - \frac{r}{2} \right) \left(z - r + \frac{2}{3} \right). \quad (3.56)$$

On $0 \leq z \leq R \leq r/2$, the maximum of E_r is therefore at 0 or R . For a noninitial crossing jump, take R to be the smaller side of its genuine starting state; the second endpoint is a fixed-smaller-side jump and is dominated by filling. For the initial crossing jump, take $R = B$, which is valid because $2B \leq B + C \leq 1 - a$. Its second endpoint is the admissible initial slab ending at (B, B) . If the zero endpoint wins, every earlier term still lies in the nonpositive part, so the empty horn dominates. Thus the crossing jump either disappears or is replaced by an admissible one-slab construction.

Merging jumps before the turn. All remaining jumps lie on the fixed- C branch. Retain the first jump, including any mandatory initial slack, and fill every later jump whose gain in (3.53) is nonpositive. If a later jump of length e has positive gain, then either $C \geq 2/3$, or

$$e > \frac{3}{2} \left(\frac{2}{3} - C \right) > \frac{2}{3} - C.$$

Equation (3.54) shows that merging it into the first jump cannot decrease the total gain. The individual gains depend only on jump lengths, so the jump and continuous lengths may be reordered along the same branch before merging. Their total length, the continuous baseline, nestedness, and the initial cap constraint are unchanged. Repetition leaves a single initial jump followed by a boundary profile.

If that jump ends at effective caps (x, y) , its contribution together with the optimal continuation is $F_1(a; x, y)$. The effective caps lie inside (B, C) , so (3.49) proves the lemma. $\square$

3.3.4 The joint three-horn inequality

Lemma 3.13 (Joint envelope). *If $a, b, c \geq 0$, $S = a + b + c \leq L$, then*

$$U_L(a; b, c) + U_L(b; a, c) + U_L(c; a, b) \leq abc \left(2L - \frac{3}{2}S \right). \quad (3.57)$$

Proof. For $x \leq y$, first omit the boundary-tail gain from (3.41) and write

$$H_L(a; x, y) = P_L(x, y) + \frac{3a - L}{2} xy(L - a - x - y). \quad (3.58)$$

Let $r = L - a - x - y$. Direct differentiation gives

$$(H_L)_y = \frac{xr}{2} [3(a + y) - L]. \quad (3.59)$$

The gain in (3.41) is active only when $y + 2x > L$. Since $x \leq y$, activity implies $y > L/3$, and hence $3(a + y) - L > 0$. Thus (3.59) is nonnegative wherever the nondecreasing gain is active. Before activation, the displayed derivative changes sign at most once, from negative to positive. Therefore, for fixed x , $F_L(a; x, y)$ first decreases and then increases as y runs from x to its allowed cap; its maximum is at the diagonal or at the full larger cap. If $a + x \geq L/3$, (3.59) sends the diagonal endpoint to the full cap. If $a + x \leq L/3$, the gain vanishes on the diagonal segment and

$$\frac{d}{dt} H_L(a; t, t) = -6t \left(t - \frac{L - a}{2} \right) \left(t - \left(\frac{L}{3} - a \right) \right) \leq 0 \quad (3.60)$$

for $0 \leq t \leq x$. That diagonal endpoint is at most zero, which is attained by the empty horn. Thus, for $B \geq C$,

$$U_L(a; B, C) = \max_{0 \leq x \leq C} F_L(a; x, B). \quad (3.61)$$

Relabel the central sides so that $a \leq b \leq c$, and first take the saturated case $L = S$. The gain in (3.41) vanishes for the horns based at b and c . For the gain-free expression, one also has

$$2(H_L)_x = (a - x)(x - y)^2 + r[x^2 + 2ay - yr], \quad (3.62)$$

where a in this formula denotes the horn's base and y its full larger cap. Equation (3.62) is nonnegative in the two relevant ranges. Consequently the b - and c -horns take their full smaller cap a . Only the horn based at the smallest side remains to be optimized.

Indeed, for the b -horn one has $0 \leq x \leq a \leq b$, larger cap c , and $r = a - x$; for the c -horn the larger cap is b and the same relations hold. Both terms on the right of (3.62) are then nonnegative.

For $0 \leq x \leq b$, set

$$\Delta = \frac{abcS}{2} - F_S(a; x, c) - H_S(b; a, c) - H_S(c; a, b). \quad (3.63)$$

Three nonnegative polynomial identities cover its entire domain. In the gain-free range $x \leq a$, substitute

$$x = p, \quad a = p + q, \quad b = p + q + r, \quad c = p + q + r + s.$$

Then

$$\begin{aligned} \Delta &= \frac{3}{4}p^2(q+r)^2 + \frac{1}{2}p(q+r)^2(s+r+2q) \\ &\quad + q^2 \left[\frac{(s+2r)^2}{4} + \frac{2}{3}q(s+2r) + \frac{1}{2}q^2 \right] \geq 0. \end{aligned} \quad (3.64)$$

In the remaining gain-free range $a \leq x \leq (a+b)/2$, substitute

$$a = p, \quad x = p + q, \quad b = p + q + r, \quad c = p + q + r + s.$$

Then

$$\begin{aligned} \Delta &= \frac{3}{4}p^2r^2 + \frac{1}{2}pr^2(s+r+2q) + \frac{1}{2}qr^2(s+r) \\ &\quad + q^2 \left[\frac{(s+2r)^2}{4} + \frac{r^2}{4} \right] + \frac{1}{3}q^3(s+2r) + \frac{1}{6}q^4 \geq 0. \end{aligned} \quad (3.65)$$

Finally, the gain is active exactly when $x \geq (a+b)/2$. Substitute

$$a = p, \quad x = p + 2q + r, \quad b = p + 2q + 2r, \quad c = p + 2q + 2r + s.$$

After including the gain in F_S , identity (3.63) becomes

$$\begin{aligned} \Delta &= \frac{3}{4}p^2r^2 + \frac{1}{2}pr^2(s+3r+4q) \\ &\quad + s^2 \left(\frac{1}{4}r^2 + qr + \frac{1}{2}q^2 \right) \\ &\quad + s \left(\frac{11}{6}r^3 + 7qr^2 + 7q^2r + \frac{7}{3}q^3 \right) \\ &\quad + \frac{31}{12}r^4 + \frac{34}{3}qr^3 + \frac{33}{2}q^2r^2 + \frac{31}{3}q^3r + \frac{31}{12}q^4 \geq 0. \end{aligned} \quad (3.66)$$

All variables in (3.64)–(3.66) are nonnegative. These cases prove (3.57) when $L = S$.

It remains to allow central slack. Fix effective smaller caps

$$0 \leq x \leq b, \quad 0 \leq y, z \leq a,$$

for the horns based at a, b, c , respectively; by (3.61), their larger caps are c, c, b . First omit the possible gain and define

$$G(L) = abc \left(2L - \frac{3}{2}S \right) - H_L(a; x, c) - H_L(b; y, c) - H_L(c; z, b). \quad (3.67)$$

For $L = S + \delta$, direct expansion gives

$$G(S + \delta) = G(S) + \delta\Lambda + \frac{\delta^2}{2}(cx + cy + bz), \quad (3.68)$$

where

$$\begin{aligned} \Lambda &= 2abc - f_a(x) - f_b(y) - f_c(z), \\ f_a(t) &= \frac{t^3}{6} + \frac{ct^2}{2} + c(a-b)t, \\ f_b(t) &= \frac{t^3}{6} + \frac{ct^2}{2} + c(b-a)t, \\ f_c(t) &= \frac{t^3}{6} + \frac{bt^2}{2} + b(c-a)t. \end{aligned} \quad (3.69)$$

These three functions are convex on their intervals; f_b, f_c are nondecreasing, while f_a is maximized at 0 or b . At those two endpoint choices, respectively,

$$\begin{aligned} \Lambda &\geq a^2 \left(\frac{b+c}{2} - \frac{a}{3} \right) \geq 0, \\ \Lambda &\geq \frac{(b-a)^2[2(b-a) + 3(c-b)]}{6} \geq 0. \end{aligned} \quad (3.70)$$

Thus the gain-free deficit is nondecreasing in L . At $L = S$, it is nonnegative for every x, y, z : equations (3.64)–(3.66) handle the largest possible b - and c -horns, whose full smaller cap is a .

Only the horn based at a can have the extra gain from (3.41), because $c + 2y \leq S$ and $b + 2z \leq S$ for the other two. Its gain

$$g((c + 2x - L)_+, c - x)$$

is nonincreasing in L . Subtracting it preserves the monotonicity of the full deficit. Since the saturated deficit is nonnegative by (3.64)–(3.66), it remains nonnegative for $L \geq S$. Maximizing the three horns independently proves (3.57). $\square$

Theorem 3.14 (Continuous no-corner inequality). *Every positive-volume box-horn set K of height one satisfies*

$$\boxed{\frac{1}{\text{vol } K} \int_K (x_1 + x_2 + x_3) dx \leq \frac{2}{3}.} \quad (3.71)$$

Proof. Apply Lemma 3.12 to the three horns and then Lemma 3.13 with $L = 1$. Their total contribution to (3.37) is at most

$$abc \left(2 - \frac{3}{2}(a + b + c) \right),$$

which is the negative of the central contribution (3.39). Hence $\mathcal{J}(K) \leq 0$, which is (3.71). $\square$

3.3.5 Application to the thickening

The rectangular thickening of Section 2.4, after positive diagonal scaling, has the box-horn form proved in Section 3.1. Its simplex containment and the preceding theorem give its deficit at least one third. Here a, b, c denote box side lengths.

3.3.6 The degree-four cardinality bound

Lemma 3.15. *If T has no full-support minimal exclusion and every point of T has total degree at most four, then*

$$|T| \leq 29. \quad (3.72)$$

Proof. Use the Section 3.1 decomposition with central point $q = (q_1, q_2, q_3)$. Necessarily $|q|_1 \leq 4$. At horn i , level $t > q_i$, its rectangular section has caps $r \leq q_j$, $s \leq q_k$ and $t + r + s \leq 4$. Consequently

$$|T| \leq \prod_{i=1}^3 (q_i + 1) + \sum_{i=1}^3 \sum_{t=q_i+1}^4 \max_{\substack{0 \leq r \leq q_j, \\ 0 \leq s \leq q_k, \\ t+r+s \leq 4}} (r+1)(s+1). \quad (3.73)$$

Dropping the nesting relation between successive sections only enlarges this upper bound. The expression is symmetric in q , so the possible sorted central triples and their values in (3.73) are:

q	Bound in (3.73)	q	Bound in (3.73)
(0, 0, 0)	13	(0, 0, 1)	19
(0, 0, 2)	23	(0, 0, 3)	25
(0, 0, 4)	25	(0, 1, 1)	25
(0, 1, 2)	28	(0, 1, 3)	28
(0, 2, 2)	28	(1, 1, 1)	29
(1, 1, 2)	29		

These are all partitions into at most three parts of an integer at most four. The largest entry is 29, proving (3.72). $\square$

4 Exact finite verification

The accompanying repository provides source code and certificates for the seven finite assertions in Section 2. Their domains and acceptance inequalities are stated in the corresponding finite lemmas. The computations enumerate a bounded generator family and two families of plane profiles, evaluate the finite degree strip, and establish three interval bounds on real parameter regions.

Each assertion is verified by two separate implementations using different representations or evaluation strategies. They independently generate the relevant families and recompute the required bounds using exact integer or rational arithmetic. The two explicit short-corner witnesses are also checked directly.

The interval certificates describe finite subdivisions of closed parameter regions. After the mathematically justified tightening from Section 2, each terminal region is proved empty, handled by an applicable analytic estimate, or bounded by a complete evaluation of the horn recurrence. The checkers verify the subdivision's coverage, including shared boundaries, and recompute every terminal bound. The certificates establish bounds throughout the closed real parameter regions.

A complete run of both implementations has verified all seven finite assertions. The repository documentation gives execution instructions, algorithm descriptions, arithmetic safeguards, and the complete-run record.

5 Extensions to embedding dimension greater than four

The moment identities extend to arbitrary embedding dimension. We give a conditional mesh-transfer principle and identify an obstruction to extending the three-horn decomposition.

5.1 Dimension-independent moment and collision identities

Let $e = d + 1$, $S = \langle m, a_1, \dots, a_d \rangle$, and define A, w, H, T, s as before, with $T \subseteq \mathbb{N}^d$. Preferred factorizations again give $|T| = m$, lower closure, and distinct residues. For the indexed family of coordinate lines, the same progression sums give

$$\sum_{\ell} L_{\ell} = dm, \quad \sum_{\ell} L_{\ell} t_{\ell} = (d + 1)s.$$

Consequently Zhai's weighted downset baseline [Zha11] reads

$$D_e := dmH - (d + 1)w \cdot s \geq 0.$$

The genus formula and $W_e = en - c = (e - 1)c - eg$ give the exact identity

$$mW_e = AD_e - \frac{e - 2}{2} m(m - 1). \quad (5.1)$$

The predecessor-subtraction collision proof also remains valid. A minimal excluded point has support disjoint from that of its representative; two distinct minimal exclusions with intersecting support have distinct residues. In particular there is still at most one full-support corner, with representative zero. Unlike in three exponent coordinates, the remaining corners can have support three or more.

Finally, if nonnegative masses on T have total dm and moment $(d + 1)s + \beta$, with $\beta \geq 0$, their destinations have weight at most H , and $w_i \geq 1$, then

$$D_e = \sum_x \alpha_x (H - w \cdot x) + w \cdot \beta \geq |\beta|_1.$$

Thus the local centroid mechanism extends. What is missing is a uniform surplus of the required size for every relevant ideal.

5.2 A conditional mesh-transfer principle

The translated-mesh argument has a useful dimension-independent form. Let K be a measurable positive-volume downset in the unit simplex. Choose an integer $N > d$, put $h = 1/N$, and define $T_z = \{y \in \mathbb{N}^d : z + hy \in K\}$. Suppose that, for almost every $z \in [0, h)^d$, this induced discrete ideal satisfies

$$dr_z |T_z| - (d + 1) \sum_{y \in T_z} |y|_1 \geq \alpha |T_z|, \quad r_z = \left\lfloor \frac{1 - |z|_1}{h} \right\rfloor.$$

Only allowances $N - d, \dots, N - 1$ occur almost everywhere. Writing $\epsilon_z = 1 - |z|_1 - hr_z$, the integrand expands as

$$\begin{aligned} \sum_{y \in T_z} (d - (d + 1)|z + hy|_1) &= hD_{r_z}(T_z) + (d\epsilon_z - |z|_1)|T_z| \\ &\geq (\alpha h - |z|_1)|T_z|. \end{aligned}$$

After integrating, each coordinate remainder has mean at most $h/2$ by the interval-fiber calculation in Section 2.11. Hence

$$d - (d + 1)\mathbb{E}_K |X|_1 \geq \left(\alpha - \frac{d}{2} \right) \frac{1}{N}. \quad (5.2)$$

Our strip transfer is $d = 3$, $N = 21$, $\alpha = 4$, giving $5/42$. The resulting bound is positive when $\alpha > d/2$.

5.3 Why the three-horn decomposition does not extend directly

The compatibility graph is chordal in three coordinate classes because an induced cycle of length at least four repeats a class. With four classes, that argument fails, and the failure is real.

Consider the finite lower ideal in $\mathbb{N}^4$

$$U = \{x \in \{0, 1\}^4 : x_1x_3 = 0, x_2x_4 = 0\}.$$

Its minimal exclusions are $2e_i$ for $1 \leq i \leq 4$, and $e_1 + e_3, e_2 + e_4$. None has full support. The compatibility graph on the four positive axis levels has the induced cycle

$$(1, 1) - (2, 1) - (3, 1) - (4, 1) - (1, 1),$$

whose two diagonals are absent.

More directly, no central box $[0, c]$, $c \in U$, together with four horns restricted to exceeding only one central coordinate can cover U . The support of c is empty, a singleton, or an adjacent pair of this cycle. In every case there is an adjacent pair disjoint from that support. Its vector $e_i + e_j$ lies in U , exceeds two central caps, and belongs to neither the central box nor any such horn.

This is an obstruction for arbitrary lower ideals, not an assertion that U admits an Apéry labeling. A higher-dimensional proof might exploit additional arithmetic restrictions or replace the decomposition with a richer structural model. The current three-horn recurrence cannot simply be reused with more axes.

Known dimension-independent results still cover substantial regions, including $3e \geq m$ [Eli20], $c \leq 3m$ [Eli18], and $t \leq e - 1$, where t is the type [FGH87]. They do not supply the missing uniform geometric surplus for a fixed $e > 4$.

6 Related work

6.1 Weighted ideals, L-shapes, and graph structure

Zhai’s asymptotic theorem [Zha11] establishes an approximate Wilf inequality for fixed embedding dimension outside finitely many semigroups. Our argument instead seeks a uniform, exact result in embedding dimension four. Within the weighted-ideal framework of Zhai and Hellus–Rechenauer–Waldi [HRW18], the additional task is a quantitative surplus sufficient to absorb the genus correction, rather than a new construction of the exponent ideal.

Aguiló-Gost, García-Sánchez, and Llena [AGGSL15] study the number of L-shapes for semigroups of embedding dimension four. Chomicz [Cho26a] gives a geometric procedure for constructing Apéry sets in that dimension and applies it to explicit families. Our moment argument uses canonical lexicographic representatives. The sum of Apéry labels is fixed, so changing their factorizations cannot improve that sum.

The clique-tree theorem used in our central-box decomposition is classical [Gav74]. The application to three-coordinate lower ideals and the resulting moment bounds are proved explicitly in Section 3.1. This use of graph theory is different from Eliahou’s graph-theoretic criterion $3e \geq m$ [Eli20].

6.2 Known regions and fixed-multiplicity computation

Fröberg, Gottlieb, and Häggkvist [FGH87] prove $g \leq tn$. Thus $t \leq e - 1$ implies Wilf immediately. Sammartano [Sam12] proves the criterion $2e \geq m$; Eliahou [Eli20] improves it to $3e \geq m$. Eliahou’s Macaulay-theorem argument [Eli18] settles $c \leq 3m$.

The computations of Bruns et al. [BGSOW20] and Kliem–Stump [KS22] use Kunz cones and their faces. Our small-multiplicity computation instead enumerates the bounded generator box obtained from the counterexample reduction in Section 2.6.

6.3 Type and presentation bounds

Chomicz’s subsequent paper [Cho26b] bounds the cardinality $\eta(S)$ of minimal presentations in terms of the type:

$$t(S) - 11 \leq \eta(S) \leq 4t(S) + 5.$$

These bounds concern another aspect of the same Apéry geometry. They do not give a uniform bound $t \leq 3$ for four generators and therefore do not replace our large-multiplicity branches. Marashdeh [Mar26] derives type bounds involving Apéry extremal statistics and reduces Wilf’s conjecture to a further inequality.

7 How the proof was constructed

Karthik Sethuraman began this project through repeated iterations with GPT-6 Astra over a period of several weeks. Those iterations produced a complete but difficult-to-read candidate proof, accompanied by numerous computational and exploratory artifacts.

Once that candidate was available, Sethuraman iterated with GPT-5.6 Sol via Codex to break apart, simplify, rephrase, and improve the proof. These repository-based iterations decomposed the argument into small proof obligations and explicit dependencies. We reconstructed the Apéry foundations, separated the geometric branches, normalized notation and endpoints, and audited the final arithmetic. We simplified the argument, removed unnecessary conditions, and replaced computations with analytic proofs where we could. In particular, the three-plane branch became entirely analytic, the low short-corner branch was reduced to local centroid predicates and two explicit witnesses, and the degree-six branch used complete local-or-axis predicates. The mesh transfer was reduced to the three allowances that occur almost everywhere.

Literature work formed part of this reconstruction. We identified and attributed the preferred-factorization, weighted-downset, initial-ideal, and support ingredients; checked the exact published multiplicity cutoff; and assessed whether Chomicz’s results could simplify the retained argument. We also examined what survives in more exponent coordinates and where the three-dimensional structure fails.

Computer assistance has two distinct roles here. AI assisted discovery, reconstruction, exposition, and checking. Exact programs establish the seven finite assertions specified mathematically in the paper, as described in Section 4.

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